

Factorisation Theorem

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Para Inf

$\underline{X} = (X_1, \dots, X_n)$ has pmf/pdf $f(\underline{x}|\theta) : \theta \in \Theta$. $T(\underline{X})$ is a sufficient statistic for θ iff $\exists$ fns h and g s.t.

$$f(\underline{x}|\theta) = \underbrace{h(\underline{x})}_{\text{free of } \theta} \underbrace{g_{\theta}(T(\underline{x}))}_{\text{fn of } \theta \text{ and } T(\underline{x})}$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U(0, \theta)$

$$f(\underline{x}|\theta) = \underbrace{\frac{1}{\theta^n} \mathbb{1}_{(x_{(n)} \leq \theta)}}_{g_{\theta}(T(\underline{x}))} \underbrace{\mathbb{1}_{(0 \leq x_{(1)})}}_{h(\underline{x})}$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U(\theta, \theta+1)$

$$f(\underline{x}|\theta) = \underbrace{\mathbb{1}_{(\theta \leq X_{(1)} \leq X_{(n)} \leq \theta+1)}}_{T(\underline{x}) = (X_{(1)}, X_{(n)}) \text{ suff stat}}$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x|\theta) = ce^{-(x-\theta)^4}; \begin{matrix} x \in \mathbb{R} \\ \theta \in \mathbb{R} \end{matrix}$

$$f(\underline{x}|\theta) = c^n e^{-\sum_{i=1}^n (x_i - \theta)^4}$$

$$= c^n e^{-\left[\sum x_i^4 - 4\sum x_i^3\theta + 6\sum x_i^2\theta^2 - 4\sum x_i\theta^3 + n\theta^4\right]}$$

$$= \underbrace{c^n e^{-\sum x_i^4}}_{h(\underline{x})} \underbrace{e^{-\left[-4\sum x_i^3\theta + 6\sum x_i^2\theta^2 - 4\sum x_i\theta^3 + n\theta^4\right]}}_{g_{\theta}(T(\underline{x}))}$$

$$T(\underline{x}) = (\sum x_i, \sum x_i^2, \sum x_i^3)$$

Remarks: (1) Suppose T is a suff statistic of θ and $T = \phi(T_1)$ where T_1 is some statistic. Is T_1 sufficient?

Pf: $f(x|\theta) = h(x) g(T(x)) = h(x) g^*(T_1(x))$
 $[g^* = g \circ \phi]$

(2) Converse not true, if T is suff, then $\phi(T)$ may not be sufficient for θ

Ex. $T = (X_1, \dots, X_n)$ sufficient but $\phi(T) = X_i$ not suff.

(3) However, if T is suff and ϕ is one-one fn. then $T_1 := \phi(T)$ is sufficient stat (H.W.)

(4) Suppose $\mathcal{P} = \{f(x|\theta) : \theta \in \Theta\}$. suppose $\Theta_0 \subset \Theta$, T is sufficient for $\mathcal{P}$ then T is sufficient for $\mathcal{P}_0$.

(5) If $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta)$ and X_i 's cont. r.v.

$$f_{(X_{(1)}, \dots, X_{(n)})}(x_1, \dots, x_n) = \begin{cases} n! f(x_1|\theta) \dots f(x_n|\theta) & \text{if } x_1 \leq x_2 \leq \dots \leq x_n \\ 0 & \text{o/w} \end{cases}$$

$T = (X_{(1)}, \dots, X_{(n)})$ is suff. stat. for θ

$$\int_{\mathcal{B}/\mathcal{C}} f_{X/T}(x_1, \dots, x_n) = \frac{1}{n!}$$

(5.1) Cauchy: $f(x|\theta) = \frac{1}{\pi(1+(x-\theta)^2)}$

$T(X)$ is only non-trivial sufficient statistic

(5.2) Laplace: $f(x|\theta) = \frac{1}{2}e^{-|x-\theta|}$ //

(5.3) Logistic $\frac{e^{-(x-\theta)}}{\{1 + e^{-(x-\theta)}\}^2}$ //

Fisher Information (Information Sufficiency)

Sometimes we can quantify information about θ contained in $\underline{X}$

Suppose $\underline{X} = (X_1, \dots, X_n)$ has pmf/pdf $f(\underline{x}|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}$
Assume conditions on $f(\underline{x}|\theta)$ open

(1) Support $\mathcal{X} = \{x: f(x|\theta) > 0\}$ does not depend on θ

(2) $\left| \frac{\partial}{\partial \theta} f(\underline{x}|\theta) \right| < \infty \quad \forall \theta \in \Theta, \forall \underline{x} \in \mathcal{X}$

(3) Derivative w.r.t. θ and integral w.r.t. $\underline{x}$ can be interchanged, i.e.,

$$0 = \frac{\partial}{\partial \theta} \int_{\mathcal{X}} f(\underline{x}|\theta) d\underline{x} = \int_{\mathcal{X}} \frac{\partial}{\partial \theta} f(\underline{x}|\theta) d\underline{x}$$

$\mathcal{X}$ $\mathcal{X}$ known

Ex satisfying (13): $N(\mu, \sigma^2)$, $\text{Bin}(k, p)$, $\text{Poi}(\lambda)$,
 $\text{Geo}(p)$, $\text{Gamma}(\alpha, \beta)$, $\text{Beta}(\alpha, \beta)$, $\text{Cauchy}(\theta, 1)$,
 $\text{Cauchy}(0, \theta)$

Def: Fisher info about θ contained in $\underline{X}$

$$I_{\underline{X}}(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\underline{X}|\theta) \right)^2 \right] \quad S(\underline{X}, \theta) := \frac{\partial}{\partial \theta} \ln f(\underline{X}|\theta)$$

= score fn.

= rate of change in the likelihood

Ex: (1) $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$

$$I_{X_1}(\mu) = \mathbb{E} \left(\frac{\partial}{\partial \mu} \left(e^{-\frac{1}{2} \ln \sigma^2 - \frac{(x-\mu)^2}{2\sigma^2}} \right) \right)^2$$

$$= \mathbb{E} \left(\frac{x-\mu}{\sigma^2} \right)^2 = \mathbb{E} \frac{(X-\mu)^2}{\sigma^4} = \frac{1}{\sigma^2}$$

$$I_{\underline{X}}(\mu) = \frac{n}{\sigma^2} = \frac{n}{V(X_1)}$$

(2) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\lambda)$

$$I_{\underline{X}}(\lambda) = \frac{n}{\lambda} = \frac{n}{V(X_1)}$$

Properties of $I_{\underline{X}}(\theta)$

(1) $\mathbb{E} \left[\underbrace{\frac{\partial}{\partial \theta} \ln f(\underline{X}|\theta)}_{S(\underline{X}, \theta)} \right] = 0$

$$\Rightarrow I_{\underline{X}}(\theta) = V \left(\frac{\partial}{\partial \theta} \ln f(\underline{X}|\theta) \right)$$

(2) $X_i \sim f_i(x|\theta) \quad \forall i=1, \dots, n \quad X_1, \dots, X_n \text{ indep.}$

$$I_{(X_1, \dots, X_n)}(\theta) = \sum_{i=1}^n I_{X_i}(\theta)$$

$$\frac{\partial}{\partial \theta} \ln f_{\underline{X}}(\underline{X}|\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \ln f_i(X_i|\theta)$$

$$V \left(\downarrow \right) = \sum_{i=1}^n V \left(\frac{\partial}{\partial \theta} \ln f_i(X_i|\theta) \right) \quad \square$$

$$(3) I_X(\theta) = - \mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(X|\theta) \right] \text{ provided}$$

$$(i) \int_{\mathcal{X}} \frac{\partial^2}{\partial \theta^2} f(x|\theta) dx = \int_{\mathcal{X}} \frac{\partial^2}{\partial \theta^2} f(x|\theta) dx$$

$$(ii) \frac{\partial^2}{\partial \theta^2} f(x|\theta) < \infty \quad \forall \theta \in \Theta \quad \forall x \in \mathcal{X}$$

$$\begin{aligned} \frac{\partial^2}{\partial \theta^2} \ln f(x|\theta) &= \frac{\partial}{\partial \theta} \left[\frac{f'}{f} \right] = \frac{f''f - f'^2}{f^2} \\ &= \left(\frac{f''}{f} \right) - \left(\frac{f'}{f} \right)^2 \\ &= \left(\frac{f''}{f} \right) - \left(\frac{\partial}{\partial \theta} \ln f \right)^2 \end{aligned}$$

Take $\mathbb{E}$ on both sides

$$\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(X|\theta) \right] = - \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(X|\theta) \right)^2 \right]$$

Information - Sufficiency

Suppose $\tilde{X}$ has pmf/pdf $f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}$ and $T(\tilde{X})$ has pmf/pdf $f_T(t|\theta)$ satisfying (1), (2), (3). Then :

$$I_{\tilde{X}}(\theta) \geq I_T(\theta) \quad \forall \theta \in \Theta$$

and "=" hold iff T is suff. stat. for θ

(C.R. Rao - Linear Statistical Inference)

Pf (for X discrete)
with 2 additional
assumptions

$$\begin{aligned} f_X(\underline{x}|\theta) &= P_\theta(X=\underline{x}) = P_\theta(X=\underline{x}, T(X)=T(\underline{x})) \\ &= \underbrace{P_\theta(X=\underline{x} | T(X)=T(\underline{x}))}_{f_{X|T}(\underline{x}|t, \theta)} \underbrace{P_\theta(T(X)=T(\underline{x}))}_{f_T(t|\theta)} \end{aligned}$$

$$\ln f_X(\underline{x}|\theta) = \ln f_{X|T}(\underline{x}|t, \theta) + \ln f_T(t|\theta) \quad T(\underline{x})=t, \text{ say}$$

$$\mathbb{E} \left(\underbrace{\frac{\partial^2}{\partial \theta^2} \ln f_X(\underline{x}|\theta)}_{I_X(\theta)} \right) = \underbrace{\mathbb{E} \left(-\frac{\partial^2}{\partial \theta^2} \ln f_{X|T}(\underline{x}|t, \theta) \right)}_{I_{X|T}(\theta)} + \underbrace{\mathbb{E} \left(-\frac{\partial^2}{\partial \theta^2} \ln f_T(t, \theta) \right)}_{I_T(\theta)}$$

$$\Leftarrow) I_{X|T}(\theta) = 0 \text{ since } f_{X|T} \dots \text{ free of } \theta$$

$$\Rightarrow) I_{X|T}(\theta) = 0 \Rightarrow \forall \left(\frac{\partial}{\partial \theta} \ln f_{X|T}(X|T, \theta) \right) = 0$$

$$\text{also } \mathbb{E} \left(\frac{\partial}{\partial \theta} \ln f_{X|T}(X|T, \theta) \right) = 0$$

$$\hookrightarrow \frac{\partial}{\partial \theta} \ln f_{X|T}(X|T, \theta) = 0$$

$$\Rightarrow f_{X|T}(X|T, \theta) \text{ free of } \theta$$

$$\Rightarrow T \text{ suff for } \theta.$$